Dear Sir or Madam,

I am a very hard-working, very talented computer programmer, who can meet the needs of your company in terms of developing any software solutions you need built. I have a mathematics degree, a little bit of professional software engineering experience, C++/Python/PHP/JavaScript, a substantial portfolio of recent programming projects, and a very good understanding of theoretical computer science and algorithms.

Because I was out of work for over ten years due to a health issue, I am currently in a position where I am turning my free time into new computer programming skills—my aspiration is, to learn one new skill per week (starting with JavaScript next week) from internet tutorials, and add more and more items to my portfolio until I am finally hired for a job as a software developer. I am determined to succeed in my software job search.

I boast a background of prestigious honors and acceptances, especially when I was a younger student; I am a little bit like John Nash, except that in addition to mathematics research, I am also extremely interested in computer programming. I excelled in math and programming as a child, taking both my first CS class and my first college-level calculus class in 8th grade. In college, I was interested in math research and game development, and took 6 computer science classes, in addition to completing the math major.

Please don't hesitate to email me (I'm not always by my phone unfortunately) to setup a phone interview! I am eager to start work in software again soon, and I hope that your company will take an interest in my capabilities and my career.

Thanks!!

Best regards,

Philip White

Resumé for Philip White
Software Developer
Email: philipjwhite2025@gmail.com
Phone: 832-417-3787

Professional Summary:

Very talented and experienced computer programmer; please check out my GitHub portfolio for evidence of this claim. Years of experience doing independent programming and also math research projects. Have been writing relatively sophisticated computer programs since the age of 12 (starting in C++ in secondary school), and had used simpler programming technologies such as PC Logo and Visual Basic since around the age of 8. Selected as a youth for membership and admission to: Center for Talented Youth, Distance Learning Program, Study of Exceptional Talent, Thomas Jefferson High School for Science and Technology. High school SAT scores: 760 verbal, 720 math. Great knowledge of undergraduate computer science and undergraduate math—see skills and knowledge section below. Eager to learn new things as an ongoing lifelong learner via self-education from advanced STEM textbooks. Informally tutored other college students at computer science and math during college.

Education:

B.S. in Mathematics
University of Mary Washington, 2008

Online Course in "Business Strategy: Competitive Advantage"
University of Pennsylvania, 2026

Online Course in "Retail Fundamentals"
Dartmouth College, 2026

Skills and Knowledge:

CS Skills

- **Please view my portfolio, at GitHub.com/cplxphil ! I have six great projects that exhibit my skills in C++ and Python in particular.**
- PHP and mySQL
- Python
- C++
- JavaScript
- HTML/CSS
- Claude Code AI to improve efficiency as a coder
- Currently, while searching for a CS job: **I am committed to learning *one new skill per week* and expanding my portfolio as much as I can before I am successfully hired.**
- Six undergraduate computer science classes, including Software Engineering
- Theoretical computer science (complexity theory, computability theory, Turing machines, descriptive complexity, finite automata)

Other skills

- Undergraduate math concepts, including: Taylor series, Markov chains, Linear regression, Statistics and probability theory, Turing machines, Computational complexity, Modular arithmetic and other concepts in number theory, Real analysis, Abstract algebra, Discrete mathematics, Graph theory, Game theory, Mathematical logic, Multivariable calculus, Polar/spherical/cylindrical coordinates, Explaining math clearly
- Other science skills: The five-factor model, economics (3 college courses), classical mechanics
- Learning styles (including abstract random, abstract sequential, concrete random, concrete sequential learning styles)
- Creative and technical writing (I took 5 English classes in college)
- Excellent interpersonal skills
- Continuing education coursework in: Retail Fundamentals, Business Strategy and Competitive Advantage
- Avid reader, especially of non-fiction textbooks, including math/science textbooks, such as, "Microeconomic Theory" by Mas-Colell, Whinston, and Green.

Programming Portfolio Projects:

1. Interactive chess bot
   - Used advanced Python object-oriented concepts, including multiple inheritance, to rapidly build a highly complex and sophisticated chess bot that can analyze all of the legal moves in any chess position and even look one move deep to analyze immediate chess material consequences of any move by the bot.
   - Also enabled human-to-human play in the chess game.
   - Proved capabilities as a backend developer, manipulating complex software mechanics and requirements to build working software that does advanced calculations and other tasks.
   - Experimented with Claude, an AI, to compile the Python source code to C++.
   - Proved skill with: Object-oriented Python, multiple inheritance, legal-move generation, recursive analysis, and C++.
2. Harvard course search HTML page with CSS
   - Demonstrated excellence in front-end web design.
   - Completed one project independently while auditing the online Harvard University Web Programming in Python/Djano with JavaScript.
   - Proved skill with: Python, HTML, CSS.
3. Staggered Stock Price change software
   - Developed a simple tool to analyze stock prices in terms of other earlier stock prices.
   - Used Python and read in from and out to files pulled from historical stocks data repositories on the internet.
   - Demonstrated advanced comprehension of mathematical and statistics concepts such as correlation and linear regression.
   - Proved skill with: Mathematics, statistics, file manipulation, lumpy, Python.
4. Playing Card Fight game
   - Creatively designed a totally new computer game using playing cards as pieces that are moved around.
   - Did excellent and serious debugging work with ChatGPT assisting me at finding glitches in the code, and, teaching me how to do parallel processing in C++, which sped up the game play significantly.
   - Demonstrated highly advanced math and CS concepts understandings, by reading online resources explaining Dijkstra's shortest path algorithm; I coded that algorithm and the rest of the AI of the game using simple, reliable, and highly safe functionalities of C++
   - Learned a new skill—SFML, the graphics library—while completing this project
   - After completing this project, I knew I was good enough at coding that I could probably code, e.g., an entire 3-D boxing came in C++ with SFML if I wanted to.
   - Proved skill with: C++, Dijkstra's algorithm, parallel processing, SFML, AI, debugging, and data structures
5. "Chat Rooms" app beginning
   - Developed a simple HTML template website for a "social chat rooms" app; I had built an app like this before on a previous computer in PHP/mySQL, but I lost access to that computer. This social chat rooms site is intended to mimic the look and functionality of the lost public chat website that I had built.
   - Proved skill with: web design, HTML, CSS, creativity.
6. JavaScript Next Prime calculator
   - To demonstrate my abilities with advanced front-end development, I created a website that uses the Rabin-Miller probabilistic algorithm for primality testing of integers.
   - Created dynamic, instant-updating code that allowed users to view the progress of the algorithm looking for a next-higher prime number in real-time.
   - Learned deep and important lessons about JavaScript and all that it can do, including the idea of "self-deleting JavaScript code" using the innerHTML properties of some elements.
   - Proved skill with: JavaScript, cryptology concepts, algorithms, mathematics.

7. Inventory Management Software
    - Demonstrated high capabilities at enterprise software by developing a complicated menus-based tree structure program for creating and assigning complex and interactive “item types” in a store.
    - Thoroughly coded a key part of the project in Python—the first menu option, and in a sense the second and third menu options, have been completed.
    - Leveraged advanced understandings of data structures such as k-ary trees (with finite natural number k) to develop sophisticated and non-trivial capabilities for managing personnel and inventory at a store.
    - Proved skill with: Python, object-oriented programming, enterprise software, file manipulation, data structures, “don’t repeat yourself” coding principles.
8. AI-built JavaScript game
    - Demonstrated excellence in understanding Claude and how to use it for well-designed JavaScript programs.
    - Proved skill with: Using AI, JavaScript, requirements communication, HTML, CSS.

Work Experience:

Substitute Mathematics Tutor
Study Smart Tutors - Part-time
June 2026 - Present - 2 months

- Selected as a qualified tutor by SST, after completing an assessment, submitting my resume and cover letter, and completing an interview in which I explained my talent at math and excellent skill at explaining mathematics to other people—I gave a brief demo of math explanations during the computer-based video interview
- Was on “stand by,” offering on call availability for substitute tutoring for middle school students in the Houston, TX, area.
- Prepared excellent math ideas to supplement the curriculum in the lessons provided by SST; delved into some philosophical concepts, in my notes preparation, regarding how to teach mathematics to anyone

Store Attendant
The Monarch Community · Part-time
Dec 2024 - Mar 2025 - 4 months, and Sep 2025 - Jan 2026 - 5 months

- Provided snack sales service to community members, making a more diverse offering of snacks available to all community members at affordable prices.
- Worked in the kitchen, washed dishes, took out trash, swept floors, and greeted diners at the community dining hall.

Data Entry Clerk
VA Department of Taxation
May 2016 - Jun 2016 - 2 months

- Provided safe and trustworthy keyboard-based data entry services to the department of taxation; complied with NDA and all other rules and guidance.

Software Engineer
Dynamics Research Corporation
Jan 2011 - Apr 2011 - 4 months

- Developed solutions in Adobe Flash for DRC to use in working with a government client.

Software Engineer
TexelTek
Jun 2010 - Aug 2010 - 3 months

- Presented highly innovative guidance/insights/ideas on government contracts to colleagues.
- Complied with NDA-based requirements to protect secret information.

Commonwealth Academy
Substitute Teacher
Mar 2010 - Apr 2010 - 1 month
- Presented lessons to high school students in the subjects of mathematics, physics, and computer science.
- Received praise from the director of the school and compliments regarding my few weeks of teaching and my knowledge of and instruction to students in computer science.

Risk Analyst
ABS Group Consulting
Feb 2009 - Oct 2009 - 9 months
- Provided consultation to ABSG regarding a Department of Homeland Security contract.
- Caught and identified a design flaw in a database application that was being designed for DHS, thereby saving the government client time and money.

Naval Surface Warfare Center Dahlgren Division
System Engineering Intern
Jun 2008 - Aug 2008 - 3 months
- Completed training in system safety engineering, while assisting the Rolling Airframe Missile (RAM) Principal for Safety with document preparation and other tasks.
- Attended lectures about cutting-edge technology, and complied with government rules regarding non-disclosure of secret information.
- Was remarked by colleagues to be quick at learning new material.

Professional References:

Janusz Konieczny
Professor of Mathematics
Academic Advisor and Mathematics Instructor
Contact email: jkoniecz@umw.edu